

DAY 16 & 17: PYTHON IMPLEMENTATION NOTE

Dynamic Manual & Super-Short Production Code Pipelines

SECTION 1: THE DYNAMIC MANUAL IMPLEMENTATION

This explicit layout maps directly to manual algebraic computations, using custom conditional logic blocks to output comprehensive analysis parameters for multi-score testing structures.

Yeh tarika hamare manual math se direct match karta hai jise ek reusable dynamic function me convert kiya gaya hai taaki aap kisi bhi production score ko daal kar check kar sakein.

```
import scipy.stats as stats

# Global Baseline Parameters
mu, sigma, alpha = 100, 10, 0.05

def run_hypothesis_test(x_score):
    # 1. Algebraic Mapping Blocks
    z_score = (x_score - mu) / sigma
    p_value = 1 - stats.norm.cdf(z_score)

    print(f"--- Score: {x_score} | Z-Score: {z_score:.2f} | p-Value: {p_value:.4f} ---")

    # 2. Logic Verification Frame
    if p_value <= alpha:
        print("    Verdict: REJECT NULL HYPOTHESIS (H0) -> REAL IMPACT!\n")
    else:
        print("    Verdict: FAIL TO REJECT NULL (H0) -> BY CHANCE.\n")

# Multi-Input Stream Testing
run_hypothesis_test(116)
run_hypothesis_test(120)
```

SECTION 2: SUPER-SHORT PRODUCTION PIPELINE

The standard industry shortcut compresses data operations directly into vectorized calculations, computing exact upper-tail probabilities in a single terminal pipeline command.

Data Science production me jab hume complex formatting nahi chahiye hoti, tab hum single line library commands ka use karke instant mathematical results nikalte hain.

```
import scipy.stats as stats

# Super-Short Vectorized Evaluation Pipeline
for x in [116, 120]:
    p = 1 - stats.norm.cdf((x - 100) / 10)
    print(f"Score: {x} | p-value: {p:.4f} | {'REJECT H0' if p <= 0.05 else 'KEEP H0'}")
```

SECTION 3: SYSTEMATIC LINE BREAKDOWN

Production Short Code Postmortem

```
for x in [116, 120]:
```

Establishes an execution loop stream allowing continuous, automated sequential scanning across any arbitrary list array of inputs.

```
for x in [116, 120]:
```

Yeh loop system ek array stream bana deta hai jisme aap jitne chahein scores ek sath bina line badle sequential checking ke liye pass kar sakte hain.

```
p = 1 - stats.norm.cdf((x - 100) / 10)
```

Combines the transformation algebra and survival distribution function directly into one vector step, mapping dynamic deviation steps directly to a p-value float vector.

```
p = 1 - stats.norm.cdf((x - 100) / 10)
```

Yeh line mathematically core calculations ko ek me merge karti hai: pehle data ka raw validation check karti hai, use normal scaling data me badalti hai aur instantly single pipeline step me exact probability value save kar leti hai.

```
{'REJECT H0' if p <= 0.05 else 'KEEP H0'}
```

Uses an inline conditional expression to evaluate critical thresholds instantly inside the print buffer output array stream.

```
{'REJECT H0' if p <= 0.05 else 'KEEP H0'}
```

Isko Python ka 'Ternary Operator' (Inline condition) kehte hain. Yeh bina lambe if-else blocks ke screen buffer ke andar hi decision terminate karke final report chipka deta hai.